5.4 Distances in a Regular Polygon

WORKSHEET

The question

STEP III 2016, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q7

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Put the polygon's vertices at the nth roots of unity, $X_k = \omega^k$. If P is the point z, what is $|PX_k|$ in terms of z? So what is the product of all the distances?

STAGE 1 The factorisation

Covers: the factorisation

Ask yourself Why is each ω^k a root of $z^n - 1$? Why are they all the roots?

STAGE 2 A point on the perpendicular bisector

Covers: Part (i)

Ask yourself Put the vertices at $X_k = \omega^k$. If P is the point z, then $|PX_k| = |z - \omega^k|$. What is the product of all the distances?

Pause and try The points equidistant from X_0 and X_1 lie on a line through O. Write P as $z = te^{i\pi/n}$ with t real. What is z^n?

Ask yourself When n is odd, treat $t > 0$ and $t < 0$ separately. What does each case mean geometrically?

STAGE 3 Chords from a vertex

Covers: Part (ii)

Ask yourself Now $P = X_0$, and one distance is zero, so $|z^n - 1|$ is no use. Which polynomial has roots $\omega, \omega^2, \ldots, \omega^{n-1}$ but not 1?

Pause and try Write that polynomial as $1 + z + \cdots + z^{n-1}$ and put $z = 1$.

What's really going on

Cotes's theorem

Part (i) is a theorem of Roger Cotes, the brilliant young Cambridge mathematician who edited the second edition of Newton's *Principia* and died in 1716 at the age of thirty-three (Newton said "if he had lived we might have known something"). Cotes found that if P lies on a ray from O through a vertex of a regular n-gon, then the product of the distances from P to the vertices is $||OP|^n - 1|$, and if P lies on a ray through the midpoint of a side, it is $|OP|^n + 1$. He used this, without complex numbers, to factorise $x^n \pm 1$ into real quadratics, which he needed for integrating rational functions. The theorem was published after his death, in 1722.

A product of sines

Since $|1-\omega^k| = |e^{-\mathrm{i}\pi k/n} - e^{\mathrm{i}\pi k/n}| = 2\sin\frac{k\pi}{n}$, part (ii) says

$$\prod_{k=1}^{n-1} \sin\frac{k\pi}{n} = \frac{n}{2^{n-1}}.$$

This is the companion of the cosine products in Section 5.3. A related product gives the *discriminant* of $z^n - 1$: the product of the squared distances between all pairs of distinct nth roots of unity is n^n. Discriminants measure how far apart the roots of a polynomial are, and they are central to the algebra of polynomials and of number fields, which we take up in Chapter 6.

Electrostatics and equal spacing

There is also a physical reading. Place n equal electric charges on a circle; the configuration that minimises the potential energy (with the logarithmic potential of two-dimensional electrostatics) is the regular polygon, and the energy involves exactly $\sum \ln|X_jX_k|$, the logarithm of a product of distances. Points that are "as spread out as possible" are important in numerical analysis (Section 3.6, where the best interpolation points come from a regular polygon via $x = \cos\theta$) and in the theory of random matrices.

Try this next

1. **Check small cases.** Verify part (ii) directly for a square ($n = 4$) and an equilateral triangle ($n = 3$).

2. **Through a vertex.** Show that if P lies on the ray from O through X_0, then $|PX_0| \times \cdots \times |PX_{n-1}| = ||OP|^n - 1|$.

3. **The average distance.** Using part (ii) and the AM–GM inequality of Section 3.2, show that the average length of the $n-1$ chords from X_0 is at least $n^{1/(n-1)}$. Find the actual average, $\frac{1}{n-1}\sum_k 2\sin\frac{k\pi}{n}$, for $n = 6$, and compare.

4. **All the chords.** HARDERShow that the product of all $\binom{n}{2}$ chord lengths of a regular n-gon inscribed in a unit circle is $n^{n/2}$.

Further reading

- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Geometry with complex numbers, beautifully illustrated.
- **Ross Honsberger, *Mathematical Gems*** (Mathematical Association of America). Several elegant problems about regular polygons.
- **Ronald Gowing, *Roger Cotes: Natural Philosopher*** (Cambridge University Press). A biography of Cotes and his mathematics.

The full guided solution is in Section 5.4:
https://beyondstep.keithmcnulty.org/ch05/polygon-distances.html